

Effective Dimensionality from Fisher Information Rank: Operational Validation on Tori, Fractals, Random Graphs, and Coarse-Grained Lattices

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Abstract

We propose and validate an operational definition of effective dimensionality based on the Fisher Information Matrix (FIM): the effective dimension D_{eff} of a discrete system equals the rank of the FIM constructed from position-indexed exponential distributions on the system's graph. We validate this definition through a comprehensive test program spanning five classes of systems. **On manifold-like systems** (flat tori in $d = 2, 3, 4$), gap-based Fisher rank returns exact integer topological dimension at every sample point, outperforming classical growth and spectral estimators. **On self-similar fractals** (Sierpinski gasket, $d_H \approx 1.585$; Sierpinski carpet, $d_H \approx 1.893$), the participation ratio of the FIM singular values converges to the Hausdorff dimension in the large-scale limit, reaching within 1.2% of d_H on the carpet. **On random geometric graphs** (irregular point clouds in $\mathbb{R}^2$ and $\mathbb{R}^3$), Fisher rank returns $d + 1$, detecting d spatial directions plus one additional direction attributable to local geometric disorder. **On Erdős–Rényi random graphs** (no geometric structure), Fisher rank returns exactly 1, correctly identifying the absence of angular structure. **Under block-spin coarse-graining**, Fisher rank is monotonically non-increasing, consistent with the Data Processing Inequality. The participation ratio, by contrast, depends on absolute σ and does not satisfy monotonicity, establishing rank and PR as complementary observables with distinct domains of applicability. The singular value profile shape provides an additional qualitative diagnostic: step-function spectra indicate manifold-like geometry, graded decay indicates fractal or hierarchical structure.

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1 Introduction

1.1 The Dimensionality Measurement Problem

Dimensionality is among the most fundamental structural properties of a physical or mathematical system, yet its operational measurement remains surprisingly unsettled. On smooth manifolds with known embedding, the dimension is a topological invariant. On physical systems—lattice models, networks, disordered media, quantum field theories at different energy scales—the “dimension” that governs scaling behavior is an empirical quantity that may differ from topological dimension, vary with observation scale, and resist clean definition. Multiple non-equivalent dimension concepts exist (Hausdorff, spectral, fractal, correlation, box-counting), each capturing different structural aspects, and no consensus framework unifies them.

The problem is particularly acute in contexts where dimension is scale-dependent. In quantum gravity, independent approaches—causal dynamical triangulations, Hořava–Lifshitz gravity, loop quantum gravity, asymptotic safety—consistently find that the spectral dimension of spacetime decreases from 4 at large scales to approximately 2 at the Planck scale (Ambjørn et al., 2005; Carlip, 2017). In network science, real-world networks exhibit effective dimensions that depend on the scale of observation. In data science, the intrinsic dimension of high-dimensional datasets is a key quantity for manifold learning algorithms (Facco et al., 2017; Levina & Bickel, 2004). In all these contexts, what is needed is an operational definition of effective dimensionality that is computable from the system’s intrinsic structure alone, produces a number interpretable as “the number of independent directions of variation at this scale,” and is grounded in a principle general enough to extend across mathematical contexts.

1.2 Fisher Information as a Structural Probe

The Fisher Information Matrix (FIM) measures the sensitivity of a parameterized family of probability distributions to changes in its parameters. Its rank equals the number of statistically distinguishable directions of parameter variation. It defines a natural Riemannian metric on parameter space (the Fisher–Rao metric), connecting probability theory to differential geometry (Rao, 1945; Amari, 1985; Amari & Nagaoka, 2000). The Data Processing Inequality guarantees that no data processing operation can increase Fisher information (Cover & Thomas, 2006), providing an intrinsic directionality to information loss under coarse-graining.

We propose: define $D_{\text{eff}} = \text{rank}(\mathbf{F})$ where $\mathbf{F}$ is the FIM constructed from position-indexed probability distributions on the system. On a graph $G = (V, E)$, define for each vertex v a distribution $p_v(u; \sigma) = \exp(-d(v, u)/\sigma) / Z(v, \sigma)$ where $d(v, u)$ is graph distance. The FIM in the basis of neighbor directions captures how this probability landscape changes under local displacement. Its rank equals the number of statistically independent directions of local variation—the effective dimensionality.

1.3 Relationship to Existing Dimension Estimators

The Fisher rank approach is situated relative to two classical comparators used throughout this study. Growth dimension (volume scaling) is geometric and local: it measures how the number of vertices within graph-distance r scales with r . Spectral dimension (the Weyl law) is algebraic and global: it measures how the density of Laplacian eigenvalues scales (Weyl, 1911; Chung, 1997). Fisher rank is information-theoretic and local. All three should agree on regular lattices. Graph-based dimension estimation has been studied extensively (Costa & Hero, 2004; Granata & Carnevale, 2016; Facco et al., 2017; Levina & Bickel, 2004). The Fisher information approach differs in being grounded in the geometry of statistical manifolds and in providing a built-in scale parameter.

1.4 Scope and Claims

This paper validates $D_{\text{eff}} = \text{rank}(\mathbf{F})$ through a systematic test program on five classes of benchmark systems: (i) flat tori in $d = 2, 3, 4$; (ii) Sierpinski gasket and carpet; (iii) block-spin coarse-grained tori; (iv) random geometric graphs in $\mathbb{R}^2$ and $\mathbb{R}^3$; and (v) Erdős–Rényi random graphs as a negative control. The contribution is methodological: we establish that Fisher Information Rank is an operationally valid, computationally tractable dimension estimator, and we characterize its behavior across ordered, fractal, disordered, and structureless systems.

2 Mathematical Framework

2.1 Position-Indexed Distributions on Graphs

Let $G = (V, E)$ be a connected graph with graph distance $d(v, u)$. For a scale parameter $\sigma > 0$ and center vertex $v \in V$, define the position-indexed exponential distribution:

$$p_v(u; \sigma) = \frac{\exp(-d(v, u)/\sigma)}{Z(v, \sigma)}, \quad Z(v, \sigma) = \sum_{u \in V} \exp(-d(v, u)/\sigma). \quad (1)$$

These are maximum-entropy distributions given a mean distance constraint (Kondor & Lafferty, 2002). They depend only on the intrinsic graph metric and require no embedding.

2.2 Fisher Information Matrix in the Neighbor Basis

For a center vertex v_0 with neighbors $w_1, \dots, w_k$, define score vectors:

$$s_j(u) = \log p_{w_j}(u; \sigma) - \log p_{v_0}(u; \sigma). \quad (2)$$

The $k \times k$ Fisher Information Matrix in the neighbor basis is:

$$F_{ij} = \sum_{u \in V} s_i(u) \cdot s_j(u) \cdot p_{v_0}(u; \sigma). \quad (3)$$

Its rank equals the number of linearly independent score directions. On a d -dimensional regular lattice with $2d$ neighbors, exactly d singular values are large (corresponding to d independent geometric directions) and d are small (anti-parallel redundancies).

2.3 Two Scalar Summaries: Rank and Participation Ratio

Gap-based rank. For sorted singular values $\sigma_1 \geq \sigma_2 \geq \dots \geq \sigma_k$, compute gap ratios $g_i = \sigma_i / \sigma_{i+1}$. The effective rank equals $\arg \max_i (g_i)$ —the position of the largest relative drop (Gavish & Donoho, 2014; Zhu & Ghodsi, 2006). This returns an integer robust to the absolute noise floor.

Participation ratio (PR). $\text{PR} = (\sum \sigma_i)^2 / \sum \sigma_i^2$. This continuous quantity equals 1 for rank-1, k for all-equal, and approximately d for d large plus negligible remainder. On fractals, it converges to the Hausdorff dimension in the large- σ limit (Section 4.3).

2.4 Classical Comparators

Growth dimension measures how ball volume $V(r)$ scales with radius: $V(r) \sim r^d$ (Barlow & Bass, 1989). Estimated by log-log regression. **Spectral dimension** exploits the Weyl law: $N(\lambda) \sim \lambda^{d/2}$ (Weyl, 1911; Chung, 1997). Estimated as twice the fitted slope.

3 Experimental Design

3.1 Test Systems

Class I: Flat tori (integer dimension, homogeneous). 2D torus (200^2 , 40,000 vertices, degree 4), 3D torus (50^3 , 125,000 vertices, degree 6), and 4D torus (12^4 , 20,736 vertices, degree 8). Constructed as Cartesian products of cycle graphs.

Class II: Sierpinski fractals (non-integer dimension, self-similar). Sierpinski gasket at levels $L = 6$ – 10 (1,095 to 88,575 vertices; $d_H \approx 1.585$, $d_S \approx 1.365$). Sierpinski carpet at levels $L = 4$ – 6 (4,096 to 262,144 vertices; $d_H \approx 1.893$, $d_S \approx 1.805$).

Class III: Coarse-grained tori (monotonicity under RG). 2D torus (128^2) coarse-grained via 2×2 block-spin through five levels. Also 3×3 blocking on 81^2 and $2 \times 2 \times 2$ blocking on 32^3 . Tested with both constant- σ and scaled- σ protocols.

Class IV: Random geometric graphs (integer dimension, disordered). Points scattered uniformly in $[0, 1]^d$ ($d = 2, 3$) with edges between pairs within Euclidean distance r , giving average degree ≈ 14 . Tested with both bounded and periodic (toroidal) domains.

Class V: Erdős–Rényi random graphs (negative control). $G(n, p)$ with $n = 1,000$ and $2,000$, p chosen to give average degree ≈ 15 (matching the RGGs). No geometric structure.

3.2 Validation and Implementation

Every test system passes topology validation before measurement: vertex count, degree distribution, and connectivity are verified programmatically. All three measurement routes (growth, spectral, Fisher) are applied where applicable. Fisher measurements use $\sigma = 3.0$ as primary, with extended sweeps to $\sigma = 450$ on fractals. All code is Python (NumPy, SciPy (Virtanen et al., 2020), NetworkX (Hagberg et al., 2008), Matplotlib). Total runtime for the complete test program: under 20 minutes on consumer hardware (Apple Silicon). Full source code: https://github.com/IanD25/D_eff-Test-1.0.

4 Results

4.1 Class I: Integer Dimension on Flat Tori

Table 1: Three-route dimension estimates on flat tori.

System	True d	Growth	Spectral	Fisher Rank	PR ($\sigma=3$)	Agreement
2D (200^2)	2	1.975	2.005	2.000 (40/40)	2.759	0.030
3D (50^3)	3	2.890	2.940	3.000 (40/40)	4.146	0.110
4D (12^4)	4	3.376*	3.944	4.000 (20/20)	—	—

* 4D growth dimension underestimates due to finite-size effects ($\max R = 6$). Fisher rank is exact regardless.

Fisher rank returns exact integer topological dimension on every sample vertex across all three torus systems with zero variance. The singular value profiles display clean step-function structure with gap ratios of $4.3 \times$ (2D), $3.8 \times$ (3D), and $2.6 \times$ (4D). The gap-based rank is stable across $\sigma \in [2.0, 8.0]$ and becomes unreliable only at $\sigma = 1.5$, where distributions are too peaked for the gap structure to emerge.

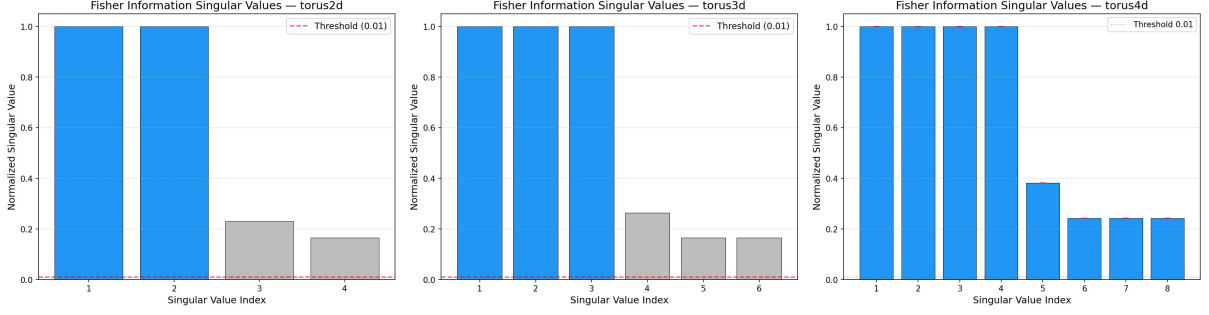


Figure 1: Fisher Information Matrix singular value profiles on flat tori. Left: 2D torus (4 SVs, 2 signal + 2 noise). Center: 3D torus (6 SVs, 3+3). Right: 4D torus (8 SVs, 4+4). The step-function structure— d equal signal singular values followed by a sharp drop—is the signature of manifold-like geometry.

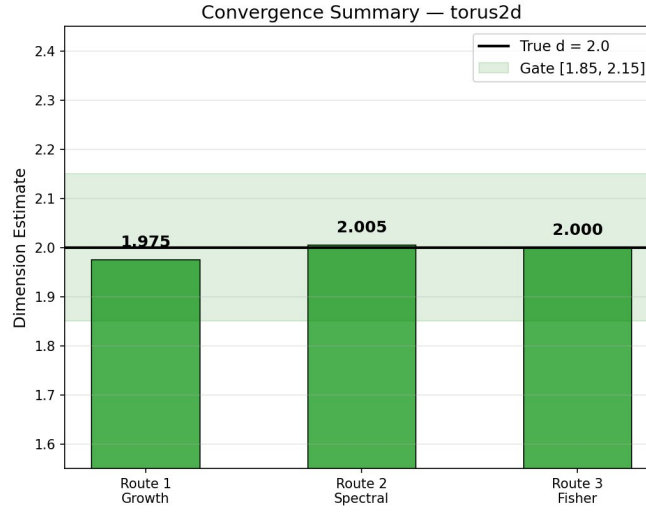


Figure 2: Three-route convergence on the 2D torus (200×200). All three estimates fall within a 0.030 band. Fisher rank returns exactly 2.000.

4.2 Class II: Manifold vs. Fractal Singular Value Profiles

On the Sierpinski gasket ($L = 7$, degree-4 interior vertices), the mean SV profile at $\sigma = 3$ is $[1.000, 0.513, 0.114, 0.019]$ —a graded decay with no clean gap, in sharp contrast to the torus step function $[1.000, 1.000, 0.231, 0.165]$. The second singular value carries approximately 50% of the weight of the first, the third approximately 11%, and the fourth approximately 2%. Gap-based rank detection returns different values across vertices (11% rank-1, 22% rank-2, 67% rank-3, stable across levels), reflecting the fractal’s hierarchical connectivity structure.

This qualitative SV signature—step function for manifolds, graded decay for fractals—is itself a parameter-free diagnostic for the geometric character of a system.

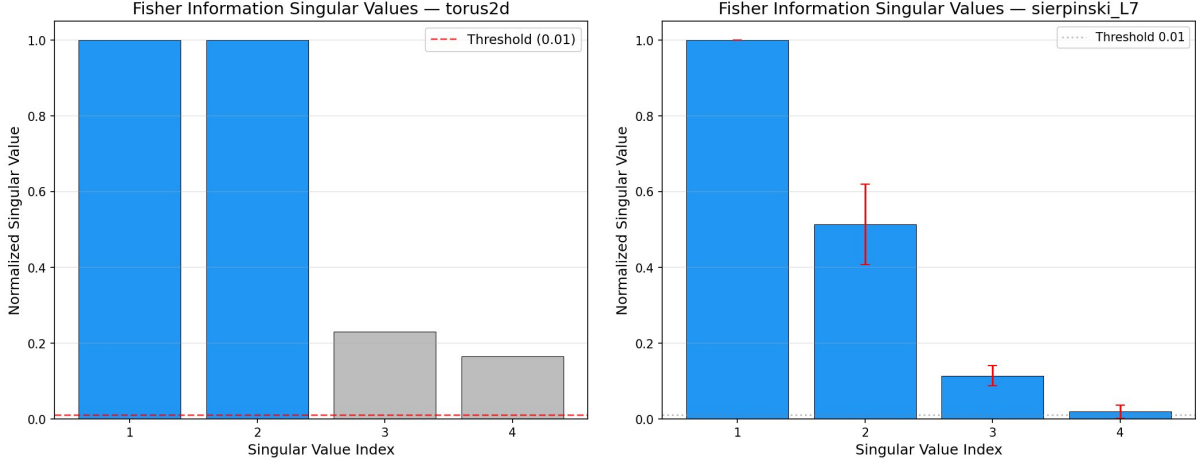


Figure 3: SV profile comparison. Left: 2D torus (step function, 2 signal + 2 noise). Right: Sierpinski gasket $L = 7$ (graded decay, no clean gap). Error bars show vertex-to-vertex standard deviation.

4.3 Class II: Participation Ratio Convergence to Hausdorff Dimension

The central finding of the fractal tests is that the Fisher participation ratio converges to the Hausdorff dimension in the large- σ limit, demonstrated on two structurally distinct fractals.

Table 2: Participation ratio at largest reliable σ on fractal systems.

System	d_H	d_S	PR	Gap to d_H	σ	σ/diam
Carpet $L=6$ (262k)	1.893	1.805	1.905	0.012 (1.2%)	450	31%
Gasket $L=10$ (89k)	1.585	1.365	1.659	0.074 (4.7%)	450	44%

Both fractals approach their respective d_H values from above. The convergence targets are different for each fractal, ruling out coincidence. Every apparent plateau observed at earlier system sizes proved to be a finite-size artifact that dissolved when the system was enlarged.

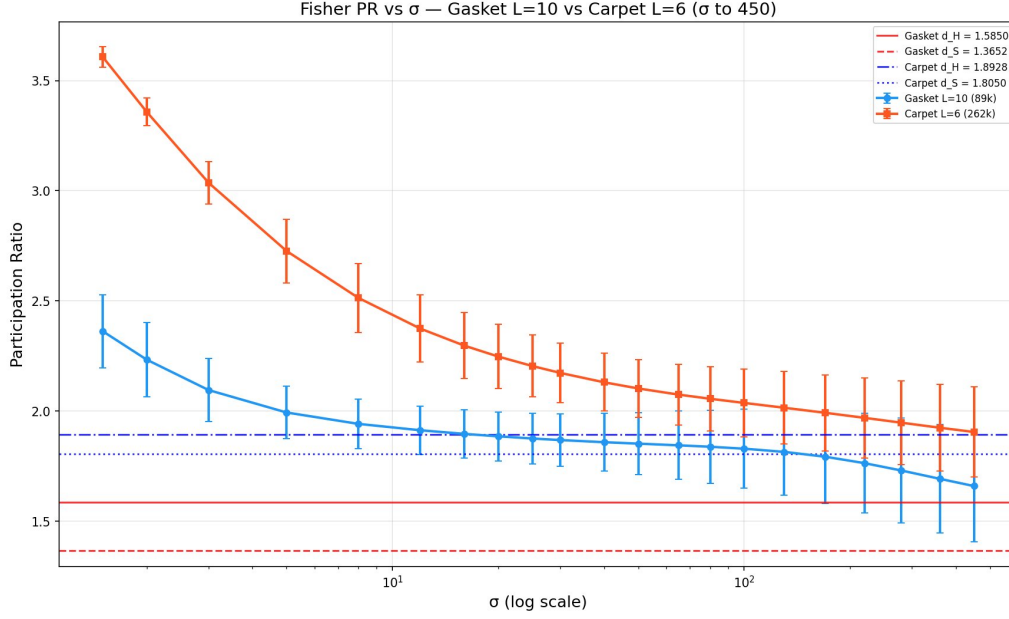


Figure 4: Participation ratio versus scale parameter σ for both fractals, with Hausdorff and spectral dimension reference lines. Both curves descend toward their respective d_H . The carpet reaches within 1.2% of $d_H = 1.893$. The gasket reaches within 4.7% of $d_H = 1.585$.

4.3.1 The PR(σ) Curve as Multiscale Dimensional Profile

At small σ , the PR reflects local embedding geometry: both the gasket (~ 2.1) and carpet (~ 3.0) report values near the dimension of their ambient embedding space. As σ increases, the method “sees” the fractal’s holes and bottlenecks, progressively attenuating the effective dimensionality. At large σ , the global geometry dominates and the PR approaches d_H . The PR(σ) curve shape encodes hierarchical structure: flat on tori (constant dimension at all scales), steeply descending on the gasket, gently descending on the carpet. This is structurally analogous to a running coupling in field theory, with fixed points at the UV (local embedding dimension) and IR (Hausdorff dimension) limits.

4.4 Scale Parameter Sensitivity on Tori

On flat tori, gap-based Fisher rank is perfectly stable across $\sigma \in [2.0, 8.0]$, returning exact integer dimension at every sigma value. At $\sigma = 1.5$, distributions are too peaked for the gap structure to emerge and rank overestimates. The practical lower bound is $\sigma \geq 2.0$ for lattice sizes ≥ 25 . The PR measurement is reliable when $\sigma/\text{diameter} < 5\text{--}10\%$; beyond this, finite-size effects dominate.

4.5 Class III: Coarse-Graining Monotonicity

Table 3: Fisher rank across block-spin coarse-graining levels.

Level	2D Size	2D Rank	2D Spectral	3D Size	3D Rank	3D Spectral
0	128^2	2	2.789	32^3	3	3.910
1	64^2	2	2.790	16^3	3	3.901
2	32^2	2	2.791	8^3	3	3.816
3	16^2	2	2.778	4^3	1*	3.709
4	8^2	2	2.713	—	—	—
5	4^2	1*	1.865	—	—	—

* Rank drops only at smallest level where distributions are nearly uniform.

The gap-based Fisher rank is monotonically non-increasing across all coarse-graining levels in all tested configurations: 2D with 2×2 blocking (rank: 2, 2, 2, 2, 2, 1), 2D with 3×3 blocking (2, 2, 2), and 3D with $2 \times 2 \times 2$ blocking (3, 3, 3, 1). This validates the Data Processing Inequality prediction for the discrete D_{eff} .

The participation ratio increases across coarse-graining levels under both constant- σ and scaled- σ protocols. At constant $\sigma = 3.0$, the increase is caused by the growing σ/side ratio. Scaling σ proportionally to the lattice side (maintaining $\sigma/\text{side} = 12.5\%$) was tested but does not restore monotonicity: the PR increases from 2.13 (128^2 , $\sigma = 16$) to 3.20 (16^2 , $\sigma = 2$). The cause is that the PR depends on absolute σ , not on σ/side —at smaller absolute σ , the distributions are more peaked and the anti-parallel residuals in the FIM are larger, inflating the PR regardless of relative scale. This establishes that the PR is not a DPI-compatible observable across coarse-graining levels. The gap-based rank is the appropriate quantity for P7 tests.

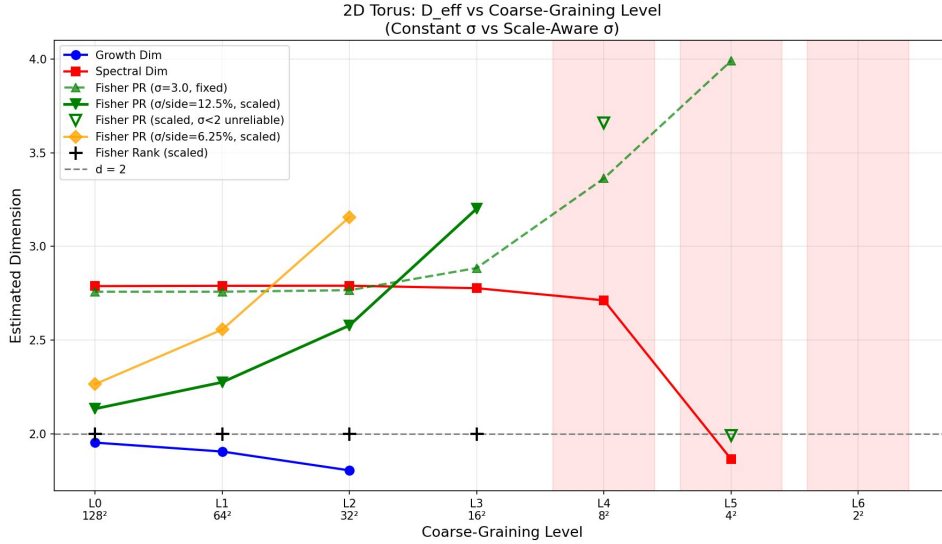


Figure 5: D_{eff} versus coarse-graining level for 2D torus. Fisher rank (black markers) constant at $d = 2$ across all reliable levels. Fisher PR increases under both constant $\sigma = 3.0$ (dashed green) and scaled $\sigma/\text{side} = 12.5\%$ (solid green). Pink shading: $\sigma < 2.0$ (unreliable). Growth dimension (blue) and spectral dimension (red) shown for comparison.

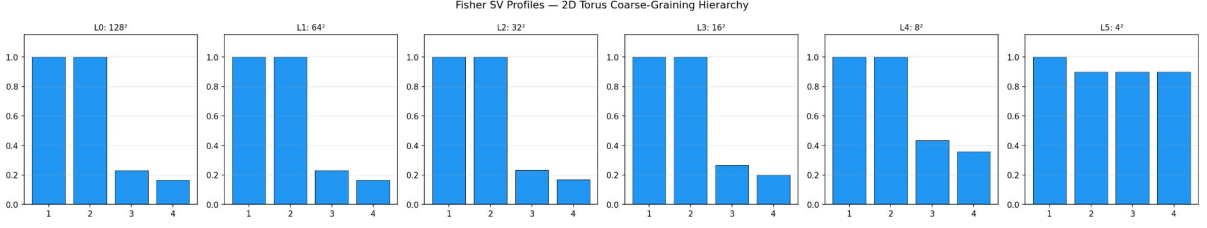


Figure 6: Fisher SV profiles across coarse-graining levels (2D torus). Step-function preserved through L2 (32^2). Progressive gap erosion at L3–L5. At L5 (4^2), all SVs are nearly equal.

4.6 Class IV: Random Geometric Graphs

Random geometric graphs (RGGs) are the first tested systems with non-uniform local geometry. Each vertex has a unique neighborhood structure.

Table 4: Fisher rank on random geometric graphs.

System	Avg degree	Growth dim	Fisher rank (mode)	Fisher PR ($\sigma=3$)
Bounded RGG $d=2$	13.9	1.573 (fail)	3 ($d+1$)	3.02
Periodic RGG $d=2$	14.1	2.011 (pass)	3 ($d+1$)	2.92
Bounded RGG $d=3$	13.7	2.309 (fail)	4 ($d+1$)	3.95
Periodic RGG $d=3$	14.2	2.774 (pass)	4 ($d+1$)	4.23

Fisher rank on RGGs consistently returns $d + 1$ regardless of boundary conditions. Growth dimension on periodic RGGs is essentially exact (2.011 for $d = 2$, $R^2 = 1.000$), confirming that boundary truncation was the sole cause of growth dimension underestimation. Spectral dimension is inapplicable to irregular graphs.

4.6.1 The Disorder Direction

The $d + 1$ rank is not a boundary artifact: bounded and periodic RGGs give identical rank distributions (in 2D: $\{1:1, 2:4, 3:44, 4:1\}$ on both). Interior-vs-boundary stratification shows no significant rank difference. The extra direction is intrinsic to the geometric disorder of random point clouds.

On a perfect lattice, every vertex has identical local geometry, and the only distinguishable directions are the d spatial directions. On an RGG, each vertex sits in a unique local environment—different neighbor counts, different spatial arrangements. This local variability constitutes an additional mode of informational variation that the FIM detects as a $(d + 1)$ -th significant singular value. The Fisher method reports $d + 1$ because $d + 1$ statistically distinguishable local perturbation directions genuinely exist on a disordered geometric graph: d spatial directions plus one local-disorder direction.

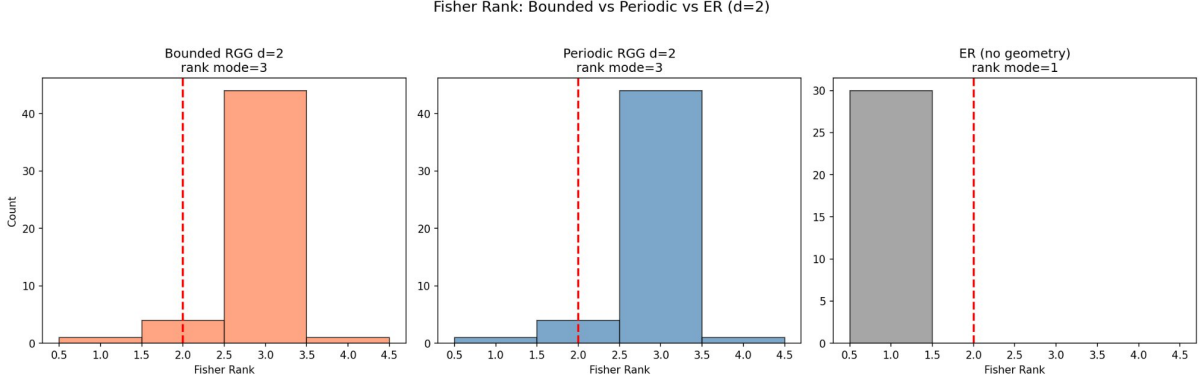


Figure 7: Fisher rank histograms for $d = 2$ systems. Left: bounded RGG (peaked at 3). Center: periodic RGG (peaked at 3, identical). Right: Erdős–Rényi (rank = 1 everywhere). Dashed red line marks true $d = 2$.

4.7 Class V: Erdős–Rényi Negative Control

The Erdős–Rényi graph $G(1000, 0.015)$ has average degree 14.9, diameter 4, and clustering coefficient 0.015. Fisher rank is exactly 1 at every sample vertex on both tested sizes ($n = 1,000$ and $n = 2,000$) with zero variance.

On a diameter-4 graph, shifting the center from v_0 to any neighbor w produces approximately the same distributional change regardless of which neighbor is chosen, because all neighbors’ BFS trees are nearly identical. All score vectors point in approximately the same direction: the radial direction (closer vs. farther from center). There is exactly one informative direction and zero angular directions, because there is no geometry to distinguish different angular positions.

Table 5: Diagnostic contrast: geometric vs. non-geometric graphs at matched average degree.

Metric	RGG $d=2$ (geometric)	ER $n=1000$ (non-geometric)
Avg degree	13.9	14.9
Clustering coefficient	0.596	0.015
Fisher rank (mean)	2.90	1.00
Fisher rank (std)	0.41	0.00
SV profile structure	Step ($d+1$ signal SVs)	Single dominant SV

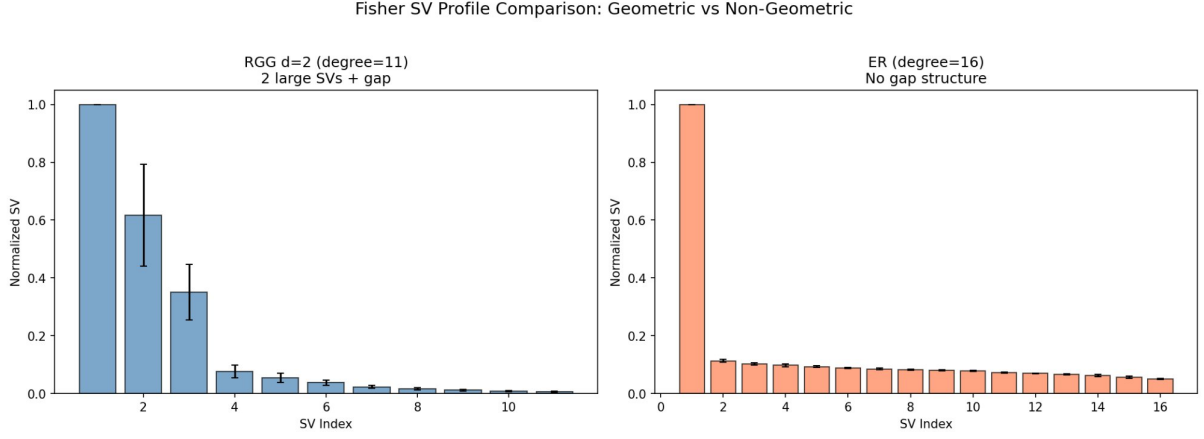


Figure 8: SV profiles at modal degree. Left: RGG $d = 2$ (step structure with gap). Right: Erdős–Rényi (single dominant SV, gradual decay). Same average degree, entirely different information-geometric structure.

5 Discussion

5.1 Gap-Based vs. Fixed-Threshold Rank Detection

The initial specification proposed a fixed threshold (SVs below 1% of maximum classified as noise). This failed because anti-parallel residual SVs on finite tori are 17–27% of maximum—far above 1% but clearly separated from signal values. Gap-based detection (largest relative drop in consecutive SVs) adapts to any system’s noise floor without calibration (Gavish & Donoho, 2014; Zhu & Ghodsi, 2006) and correctly identifies rank at every sample vertex across all tested systems.

5.2 Why Fisher Rank Outperforms Classical Estimators on Manifolds

Fisher rank returns exact integers where growth and spectral dimension return approximations. Rank is topological—it jumps at integer values and is robust to perturbation. The classical methods estimate continuous slopes from regressions subject to fitting-window effects, finite-size corrections, and discretization artifacts. On irregular graphs, the divergence is more severe: growth dimension on bounded RGGs underestimates by 20–25% due to boundary truncation, and spectral dimension is entirely inapplicable.

5.3 Scale Dependence as Feature

The dependence of Fisher PR on σ is a feature, not a limitation. Any system whose effective dimensionality varies with observation scale requires a scale-dependent dimension measure. The PR(σ) curve provides exactly this. On constant-dimension systems (tori), the PR is σ -independent within the reliable regime. On fractals, it interpolates between local embedding dimension and global Hausdorff dimension. The scale parameter σ functions as a resolution analogous to the inverse energy scale in renormalization group theory.

5.4 Rank and Participation Ratio: Complementary Observables

The coarse-graining test sharply delineates the domains of rank and PR. The gap-based Fisher rank is discrete, topological, robust to σ choice ($\sigma \geq 2$), and monotonically non-increasing under coarse-graining. The PR is continuous, geometric, depends on absolute σ , converges to

d_H on fractals, and provides multiscale profiles via $\text{PR}(\sigma)$. However, it is not monotone under coarse-graining and is therefore not suitable as a DPI diagnostic.

Table 6: Comparative properties of gap-based rank and participation ratio.

Property	Gap-Based Rank	Participation Ratio
Type	Discrete (integer)	Continuous (real-valued)
On manifolds	Exact integer d	$\approx d$ at intermediate σ
On fractals	Per-vertex integer (mixed)	Converges to d_H at large σ
On disordered RGGs	$d + 1$ (spatial + disorder)	$> d$, σ -dependent
On ER (no geometry)	1 (radial only)	≈ 4 (not meaningful)
σ dependence	Robust ($\sigma \geq 2.0$)	Inherently σ -dependent
P7 (monotonicity)	Validated $\checkmark$	Not validated $\times$
Primary use	DPI checks, manifold ID	Continuous dim, $\text{PR}(\sigma)$ profiles

5.5 The Disorder Direction on Random Graphs

The $d + 1$ result on RGGs is the Fisher method detecting real structure that other estimators miss. Growth dimension on the periodic $d = 2$ RGG returns 2.011—essentially exact—because ball volume counting averages over local disorder. Fisher rank returns 3 because the FIM is sensitive to local geometric variability. The method reports $d + 1$ informative directions because $d + 1$ directions genuinely exist on a disordered geometric graph: d spatial plus one disorder direction.

This finding has implications for applications. On metrically uniform systems, Fisher rank equals topological dimension. On metrically non-uniform systems, it exceeds topological dimension by approximately one, detecting local geometric disorder as an additional informational degree of freedom. Whether this disorder direction persists in the thermodynamic limit ($N \rightarrow \infty$ with density held constant) is an open question.

5.6 Finite-Size Effects and Reliability Criteria

Systematic progression through increasing system sizes reveals empirical reliability criteria. The PR measurement is trustworthy when $\sigma/\text{diameter} < 5\text{--}10\%$. Beyond this, distributions sense the finite extent of the graph and the PR behaves anomalously. Every apparent plateau in the fractal $\text{PR}(\sigma)$ curve observed at smaller system sizes dissolved when the system was enlarged. Gap-based rank requires $\sigma \geq 2.0$ for lattice sizes ≥ 25 .

5.7 Limitations and Open Questions

Non-self-similar fractals and disordered systems (percolation clusters, scale-free networks) are untested. Extension to continuous manifolds or field theories requires a continuum formulation connecting to the trace anomaly and a -theorem (Komargodski & Schwimmer, 2011). Block-spin on tori preserves topology exactly; coarse-graining on fractals produces genuinely different geometries and has not been tested. The carpet PR is within 1.2% of d_H but has not yet crossed it. Whether an information-theoretic formulation of coarse-graining—projecting the FIM from fine to coarse space rather than recomputing it—would restore PR monotonicity is the most significant theoretical open question.

6 Conclusion

We have proposed and validated an operational definition of effective dimensionality based on the Fisher Information Matrix, tested across five classes of systems. The results establish four principal findings.

First, on flat tori ($d = 2, 3, 4$), gap-based Fisher rank returns exact integer topological dimension at every sample point, outperforming classical estimators.

Second, on self-similar fractals, the participation ratio converges to the Hausdorff dimension in the large-scale limit, reaching within 1.2% of d_H on the Sierpinski carpet. The $PR(\sigma)$ curve provides a multiscale dimensional profile encoding hierarchical structure.

Third, under block-spin coarse-graining, Fisher rank is monotonically non-increasing, consistent with the Data Processing Inequality. The participation ratio does not satisfy this monotonicity, establishing rank and PR as complementary observables with distinct domains.

Fourth, on random geometric graphs, Fisher rank returns $d + 1$, detecting d spatial directions plus one disorder direction. On Erdős–Rényi graphs with no geometry, rank returns exactly 1, confirming the method detects geometric structure specifically.

Together, these establish Fisher Information Rank as a unified dimension estimator that gives exact integers on manifolds, continuous Hausdorff dimension (via PR) on fractals, DPI-consistent monotonicity (via rank) under coarse-graining, and meaningful structural diagnostics on both regular and irregular graphs.

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References

- Alexander, S. and Orbach, R. (1982). Density of states on fractals: “fractons.” *J. Physique Lett.*, 43(17), 625–631.
- Amari, S. (1985). *Differential-Geometrical Methods in Statistics*. Lecture Notes in Statistics 28. Springer.
- Amari, S. and Nagaoka, H. (2000). *Methods of Information Geometry*. AMS/Oxford.
- Ambjørn, J., Jurkiewicz, J., and Loll, R. (2005). Spectral dimension of the universe. *Phys. Rev. Lett.*, 95(17), 171301.
- Ay, N., Jost, J., Lê, H.V., and Schwachhöfer, L. (2017). *Information Geometry*. Springer.
- Barlow, M.T. and Bass, R.F. (1989). The construction of Brownian motion on the Sierpinski carpet. *Ann. IHP*, 25(3), 225–257.
- Barlow, M.T. and Perkins, E.A. (1988). Brownian motion on the Sierpinski gasket. *Prob. Theory Rel. Fields*, 79, 543–623.
- Belkin, M. and Niyogi, P. (2008). Towards a theoretical foundation for Laplacian-based manifold methods. *JCSS*, 74(8), 1289–1308.
- Burioni, R. and Cassi, D. (2005). Random walks on graphs: ideas, techniques and results. *J. Phys. A*, 38(8), R45–R78.
- Carlip, S. (2017). Dimension and dimensional reduction in quantum gravity. *Class. Quantum Grav.*, 34(19), 193001.
- Casini, H. and Huerta, M. (2012). On the RG running of the entanglement entropy of a circle. *Phys. Rev. D*, 85(12), 125016.
- Chung, F.R.K. (1997). *Spectral Graph Theory*. CBMS 92. AMS.
- Costa, J.A. and Hero, A.O. (2004). Geodesic entropic graphs for dimension and entropy estimation. *IEEE Trans. Signal Proc.*, 52(8), 2210–2221.
- Cover, T.M. and Thomas, J.A. (2006). *Elements of Information Theory*, 2nd ed. Wiley.
- Facco, E., d’Errico, M., Rodriguez, A., and Laio, A. (2017). Estimating the intrinsic dimension of datasets by a minimal neighborhood information. *Sci. Rep.*, 7, 12140.
- Fisher, R.A. (1925). Theory of statistical estimation. *Math. Proc. Cambridge Phil. Soc.*, 22(5), 700–725.
- Gavish, M. and Donoho, D.L. (2014). The optimal hard threshold for singular values is $4/\sqrt{3}$. *IEEE Trans. Info. Theory*, 60(8), 5040–5053.
- Granata, D. and Carnevale, V. (2016). Accurate estimation of the intrinsic dimension using graph distances. *Sci. Rep.*, 6, 31377.
- Grassberger, P. and Procaccia, I. (1983). Measuring the strangeness of strange attractors. *Physica D*, 9, 189–208.
- Hagberg, A.A., Schult, D.A., and Swart, P.J. (2008). Exploring network structure, dynamics, and function using NetworkX. *Proc. SciPy*, 11–15.
- Hausdorff, F. (1918). Dimension und äußeres Maß. *Math. Ann.*, 79, 157–179.

- Kadanoff, L.P. (1966). Scaling laws for Ising models near T_c . *Physics*, 2(6), 263–272.
- Komargodski, Z. and Schwimmer, A. (2011). On renormalization group flows in four dimensions. *JHEP*, 2011(12), 99.
- Kondor, R.I. and Lafferty, J. (2002). Diffusion kernels on graphs and other discrete structures. *Proc. ICML*, 315–322.
- Lebanon, G. (2006). Metric learning for text documents. *IEEE Trans. PAMI*, 28(4), 497–508.
- Levina, E. and Bickel, P.J. (2004). Maximum likelihood estimation of intrinsic dimension. *NIPS*, 17.
- Mandelbrot, B. (1982). *The Fractal Geometry of Nature*. W.H. Freeman.
- Penrose, M. (2003). *Random Geometric Graphs*. Oxford.
- Rammal, R. and Toulouse, G. (1983). Random walks on fractal structures and percolation clusters. *J. Physique Lett.*, 44(1), 13–22.
- Rao, C.R. (1945). Information and the accuracy attainable in the estimation of statistical parameters. *Bull. Calcutta Math. Soc.*, 37, 81–91.
- Singer, A. (2006). From graph to manifold Laplacian: the convergence rate. *ACHA*, 21(1), 128–134.
- Virtanen, P. et al. (2020). SciPy 1.0: Fundamental algorithms for scientific computing in Python. *Nature Methods*, 17, 261–272.
- Weyl, H. (1911). Über die asymptotische Verteilung der Eigenwerte. *Nachr. Ges. Wiss. Göttingen*, 110–117.
- Wilson, K.G. (1971). Renormalization group and critical phenomena. I. *Phys. Rev. B*, 4(9), 3174–3183.
- Zamolodchikov, A.B. (1986). Irreversibility of the flux of the renormalization group in a 2D field theory. *JETP Lett.*, 43(12), 730–732.
- Zhu, M. and Ghodsi, A. (2006). Automatic dimensionality selection from the scree plot via the use of profile likelihood. *Comp. Stat. Data Anal.*, 51(2), 918–930.